PAPER_669: UQFF Waveform Comparison — GW150914

Subtitle: UQFF-modified strain h_UQFF vs LIGO GW150914 observation. Inspiral chirp mass, frequency evolution, and phase shift derived. **Module:** UQFFCompared-ToGW150914
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Abstract

We compute the UQFF-modified gravitational wave strain h_UQFF for the GW150914 binary black hole merger and compare to LIGO observation. UQFF introduces a ~10% strain suppression and a small phase shift controlled by κ and f_TRZ.

1. GW150914 Parameters

- m1 = 36 M$\odot$, m2 = 29 M$\odot$
- d = 410 Mpc
- f_peak = 150 Hz, h_obs $\approx 10^{-21}$

2. Chirp Mass

$$\mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \approx 28.3\, M_\odot$$

3. Standard Strain

$$h_{GR}(f) = \frac{4}{d} \left(\frac{G\mathcal{M}_c}{c^2} \right)^{5/3} (\pi f)^{2/3}/c$$

4. UQFF Modifications

$$S_{SCm}(f) = \exp(-\rho_{SCm}\, \lambda_{GW}), \quad \lambda_{GW} = c/f$$

$$h_{UQFF}(f) = h_{GR}(f) \cdot (1 - f_{TRZ}) \cdot S_{SCm}(f) \cdot \exp\left(-\frac{U_m\, \omega}{c^2}\right)$$

5. Phase Shift

$$\phi_{UQFF}(t) = 2\pi f t + \kappa\, f_{TRZ}\, t$$

Small but potentially measurable with future detectors (Einstein Telescope).

6. Frequency Evolution

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \frac{(G\mathcal{M}_c)^{5/3}}{c^5} f^{11/3}$$

7. C++ Module

`UQFFComparedToGW150914.h` / `.cpp` — Session 172 CP4 #253 — `UQFFComparedToGW150914Calculator`

PAPER_669 | Session 172 | Star-Magic UQFF Framework v5.29 | Daniel Murphy